

Carrier 38MURA Installation Manual (2019)

Model 38MURA | 2019 | Installation

IM-38MURA-2019

Prototype sample document for the clickable demo. Not an official Carrier publication.

Rev	Date	Description
1.0	2019-05	Initial release for the 2019 model year.
1.1	2019-09	Updated CN2 pinout and fault-code table.
1.2	2020-02	Added inverter diagnostics and torque values.

> For professional use only. Service work must be performed by qualified personnel.

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> Page numbers refer to the printed page footer. Cross-references use section numbers.

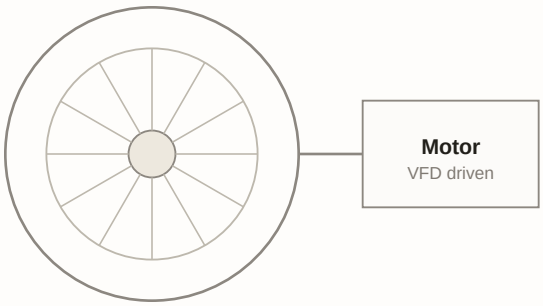
3. Receiving and Inspection

Use a calibrated multimeter with a true-RMS function for all AC measurements. Resistance checks must be made with the unit de-energized and the harness disconnected.

Procedure notes

- Measure with the unit in the same mode as the fault.
- Allow readings to stabilize for 30 seconds.
- Note the ambient temperature at the time of test.
- Repeat any suspect measurement twice.
- Compare to the published range, not to an assumption.

Step	Tool	Spec
Mounting screws	1/4 in driver	2.5 N-m
Terminals	Torque driver	0.8 N-m
Line set	Flare wrench	per chart
Filter	Hand	snug
Panel	1/4 in driver	1.5 N-m



Indoor blower wheel and drive motor
Receiving and Inspection reference

> Use the wiring diagram in Appendix D to confirm connector pinouts before probing.

Startup Sequence and Fault Codes

Section 3.1: Startup

Power the unit and confirm the board LED runs its startup sequence. The startup table lists each code and its meaning. The table gives the code meaning only; use the Service Manual for the diagnostic procedure.

1. Confirm all electrical and refrigerant connections are complete.
2. Energize the unit at the service switch.
3. Observe the board LED startup sequence.
4. Record any fault code before it clears.

Code	Meaning	Reference
E1	High pressure	Service Manual 5.3
E2	Low pressure	Service Manual 5.3
E4	Locked rotor	Service Manual 5.5
E5	AC overcurrent	Service Manual 5.2

> E5 is listed as an AC overcurrent fault in the startup table. Load the Service Manual to complete the diagnostic procedure.

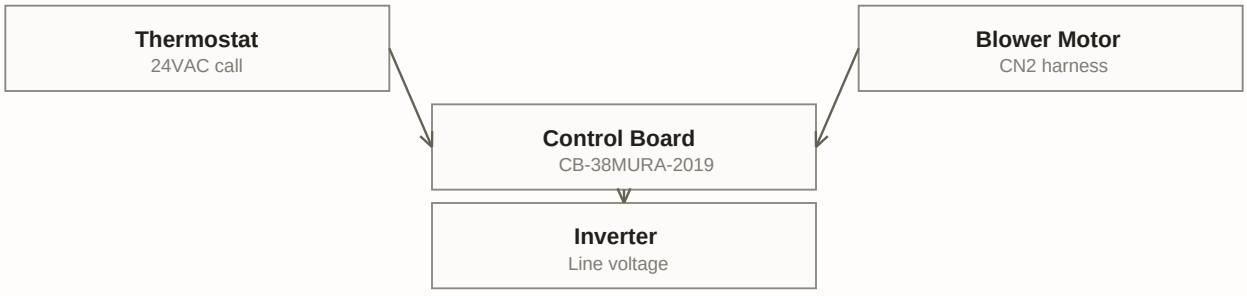
5. Startup Checklist

The board stores the last five fault codes in non-volatile memory. Review the stored codes before clearing them, because the history often points to the true root cause.

Procedure notes

- Verify thermostat demand is present.
- Check the low-voltage transformer output.
- Measure the signal at CN2.
- Compare against the table in this section.
- Clear codes only after the repair is verified.

Item	Expected	Notes
Line voltage	208-230 VAC	+/-10%
Low voltage	24 VAC	+/-10%
DC bus	310 VDC	Discharged before service
CN2 signal	24 VAC	Call for fan
Ambient	see chart	Record at test



Startup Checklist reference

> Record the outdoor ambient and the suction line temperature together for an accurate comparison.

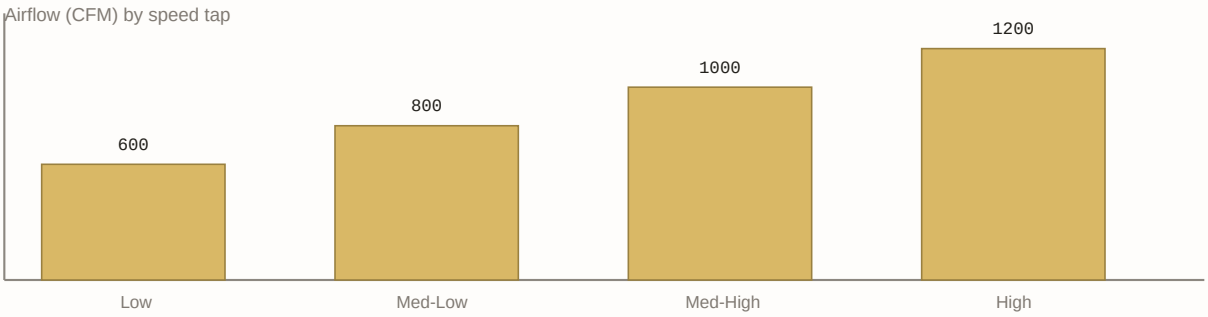
6. Mounting the Outdoor Unit

Record ambient and supply-air temperatures at the same time so the temperature split can be compared against the charging chart for the current mode.

Procedure notes

- Use only factory-approved replacement parts.
- Match the full part number, including revision.
- Transfer configuration jumpers where fitted.
- Confirm harness routing against the reference photos.
- Run a full startup sequence after assembly.

Terminal	Wire	Function
CN2-1	RED	24VAC supply
CN2-2	WHT	Common return
CN2-3	BLU	Tach feedback
CN1-1	BLK	Earth ground
CN1-2	YEL	Thermostat Y



Mounting the Outdoor Unit reference

> If a code returns after clearing, inspect the harness before replacing the board.

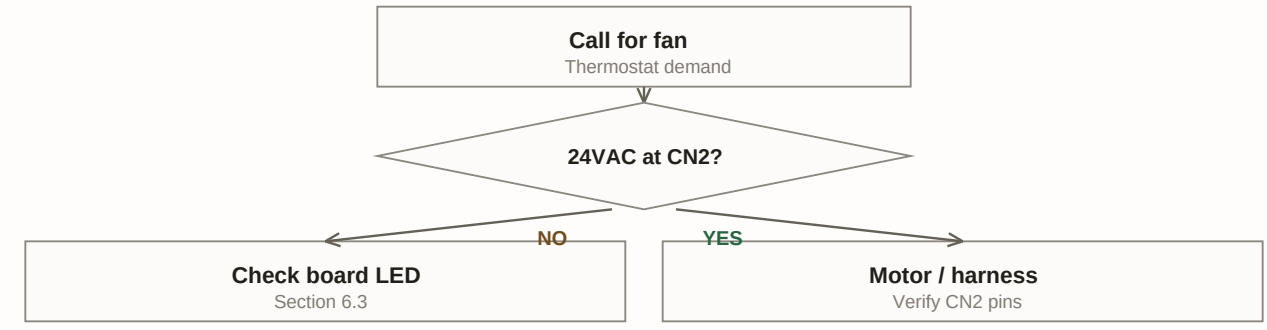
7. Clearances

When a reading is out of range, do not replace the component until the associated wiring and connectors have been verified good. Most returned parts test good on the bench.

Procedure notes

- Measure with the unit in the same mode as the fault.
- Allow readings to stabilize for 30 seconds.
- Note the ambient temperature at the time of test.
- Repeat any suspect measurement twice.
- Compare to the published range, not to an assumption.

Code	Meaning	Action
E1	High pressure	Check charge and airflow
E2	Low pressure	Check charge and leak
E4	Locked rotor	Check compressor
E5	AC overcurrent	See Section 5.2
E6	Communication	Check harness



Clearances reference

> Use the wiring diagram in Appendix D to confirm connector pinouts before probing.

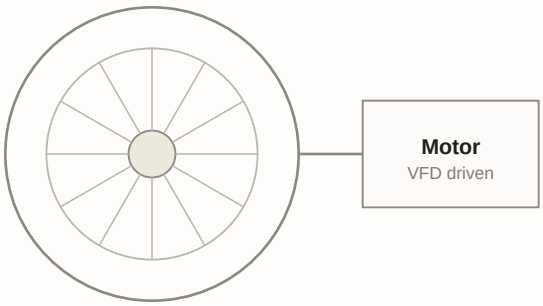
8. Refrigerant Line Sets

Torque all electrical terminations to the value shown in the specifications table. Loose terminations overheat and can damage the board terminals over time.

Procedure notes

- Disconnect and lock out power.
- Confirm 0VAC and 0VDC at the board.
- Inspect connectors for corrosion.
- Record baseline readings before any change.
- Reassemble and re-check operation.

Step	Tool	Spec
Mounting screws	1/4 in driver	2.5 N-m
Terminals	Torque driver	0.8 N-m
Line set	Flare wrench	per chart
Filter	Hand	snug
Panel	1/4 in driver	1.5 N-m



Indoor blower wheel and drive motor
Refrigerant Line Sets reference

> Always confirm 0VAC and 0VDC at the board before making resistance measurements.

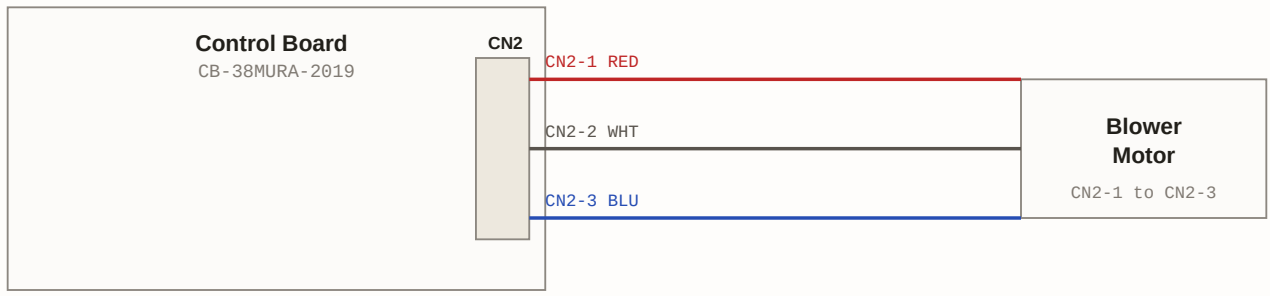
9. Line Set Sizing

Document the final readings and the parts used on the work order. Accurate records shorten the next visit and support the warranty claim.

Procedure notes

- Verify thermostat demand is present.
- Check the low-voltage transformer output.
- Measure the signal at CN2.
- Compare against the table in this section.
- Clear codes only after the repair is verified.

Speed Tap	CFM	Static
Low	600	0.20 in wc
Med-Low	800	0.30 in wc
Med-High	1000	0.40 in wc
High	1200	0.50 in wc



24VAC control signal from board to blower motor

Line Set Sizing reference

> Record the outdoor ambient and the suction line temperature together for an accurate comparison.

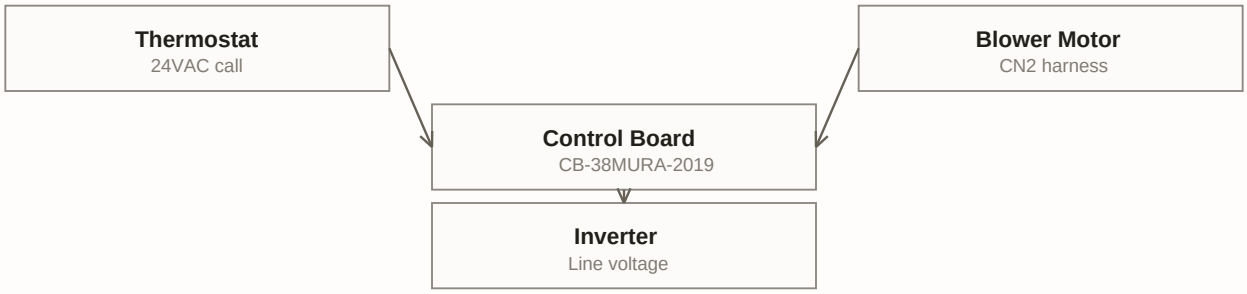
10. Brazing and Purge

This section covers the recommended field procedure for the 2019 Carrier 38MURA. Follow the steps in order and verify each measurement before moving to the next step. Record every reading on the service worksheet.

Procedure notes

- Use only factory-approved replacement parts.
- Match the full part number, including revision.
- Transfer configuration jumpers where fitted.
- Confirm harness routing against the reference photos.
- Run a full startup sequence after assembly.

Item	Expected	Notes
Line voltage	208-230 VAC	+/-10%
Low voltage	24 VAC	+/-10%
DC bus	310 VDC	Discharged before service
CN2 signal	24 VAC	Call for fan
Ambient	see chart	Record at test



Brazing and Purge reference

> If a code returns after clearing, inspect the harness before replacing the board.

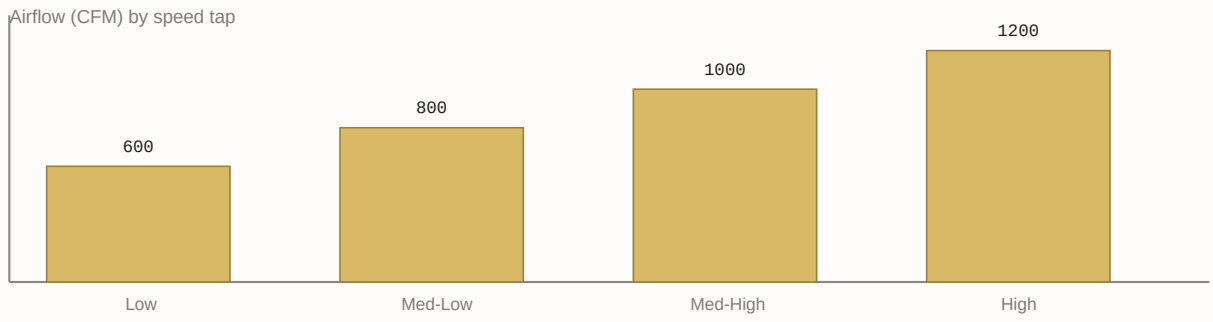
11. Pressure Testing

The 38MURA uses a variable-speed inverter to match capacity to load. Because the inverter regulates motor speed, most faults present as a control-signal problem rather than a mechanical failure.

Procedure notes

- Measure with the unit in the same mode as the fault.
- Allow readings to stabilize for 30 seconds.
- Note the ambient temperature at the time of test.
- Repeat any suspect measurement twice.
- Compare to the published range, not to an assumption.

Terminal	Wire	Function
CN2-1	RED	24VAC supply
CN2-2	WHT	Common return
CN2-3	BLU	Tach feedback
CN1-1	BLK	Earth ground
CN1-2	YEL	Thermostat Y



Pressure Testing reference

> Use the wiring diagram in Appendix D to confirm connector pinouts before probing.

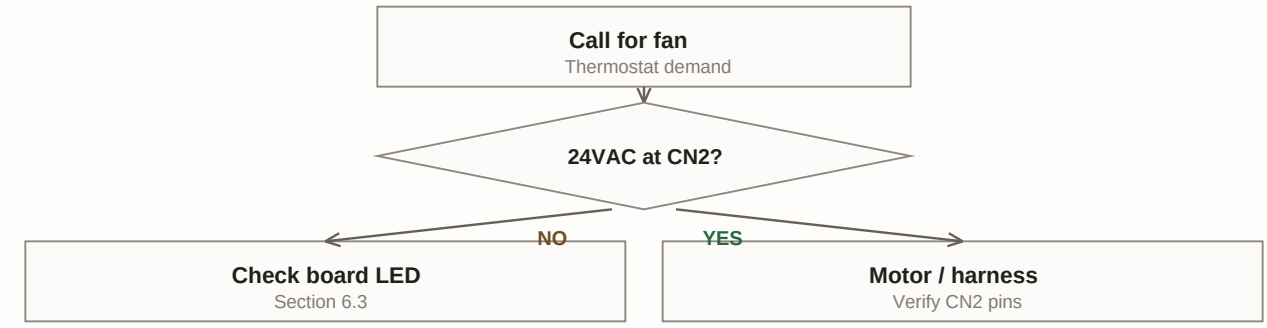
12. Evacuation

Before opening any panel, disconnect power at the service switch and wait for the DC bus to discharge. Confirm 0VAC and 0VDC at the board before proceeding.

Procedure notes

- Disconnect and lock out power.
- Confirm 0VAC and 0VDC at the board.
- Inspect connectors for corrosion.
- Record baseline readings before any change.
- Reassemble and re-check operation.

Code	Meaning	Action
E1	High pressure	Check charge and airflow
E2	Low pressure	Check charge and leak
E4	Locked rotor	Check compressor
E5	AC overcurrent	See Section 5.2
E6	Communication	Check harness



Evacuation reference

> Always confirm 0VAC and 0VDC at the board before making resistance measurements.

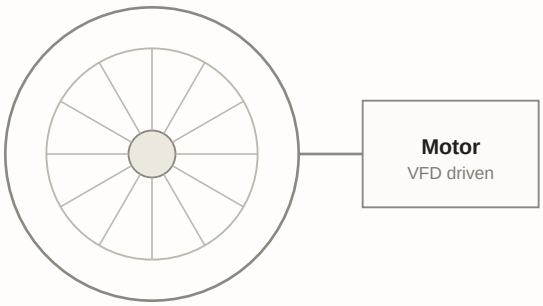
13. Electrical Service

Use a calibrated multimeter with a true-RMS function for all AC measurements. Resistance checks must be made with the unit de-energized and the harness disconnected.

Procedure notes

- Verify thermostat demand is present.
- Check the low-voltage transformer output.
- Measure the signal at CN2.
- Compare against the table in this section.
- Clear codes only after the repair is verified.

Step	Tool	Spec
Mounting screws	1/4 in driver	2.5 N-m
Terminals	Torque driver	0.8 N-m
Line set	Flare wrench	per chart
Filter	Hand	snug
Panel	1/4 in driver	1.5 N-m



Indoor blower wheel and drive motor
Electrical Service reference

> Record the outdoor ambient and the suction line temperature together for an accurate comparison.

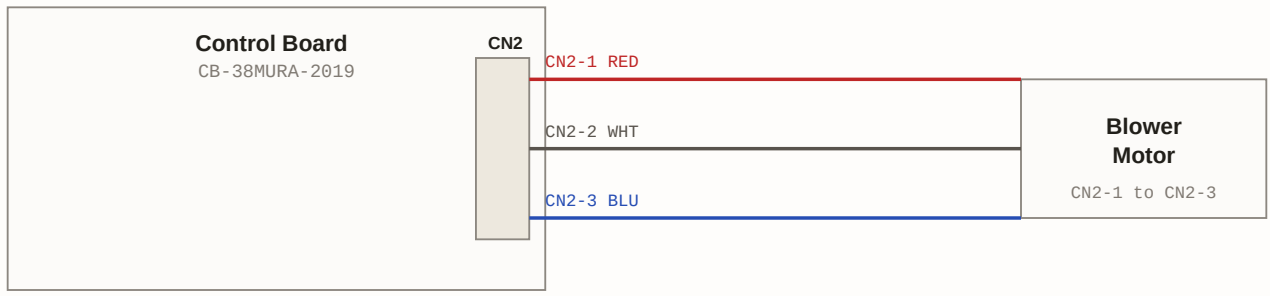
14. Low-Voltage Wiring

Connector CN2 carries the low-voltage control signal between the board and the blower motor. A loose or corroded pin here is one of the most common causes of intermittent operation.

Procedure notes

- Use only factory-approved replacement parts.
- Match the full part number, including revision.
- Transfer configuration jumpers where fitted.
- Confirm harness routing against the reference photos.
- Run a full startup sequence after assembly.

Speed Tap	CFM	Static
Low	600	0.20 in wc
Med-Low	800	0.30 in wc
Med-High	1000	0.40 in wc
High	1200	0.50 in wc



24VAC control signal from board to blower motor

Low-Voltage Wiring reference

> If a code returns after clearing, inspect the harness before replacing the board.

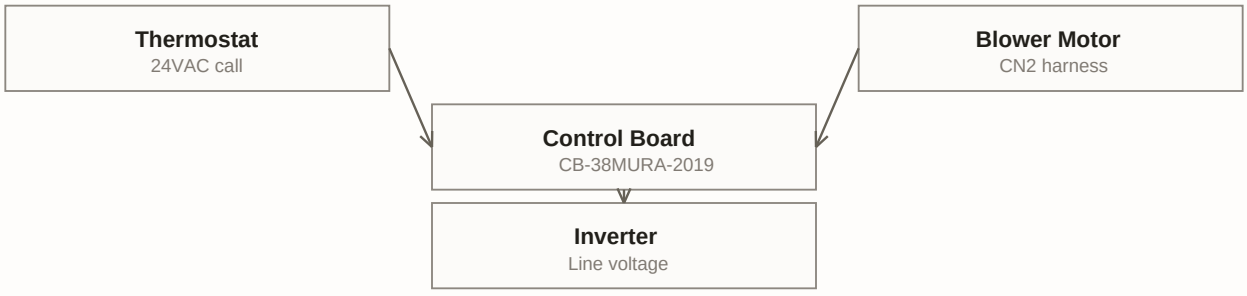
15. Thermostat Wiring

The board stores the last five fault codes in non-volatile memory. Review the stored codes before clearing them, because the history often points to the true root cause.

Procedure notes

- Measure with the unit in the same mode as the fault.
- Allow readings to stabilize for 30 seconds.
- Note the ambient temperature at the time of test.
- Repeat any suspect measurement twice.
- Compare to the published range, not to an assumption.

Item	Expected	Notes
Line voltage	208-230 VAC	+/-10%
Low voltage	24 VAC	+/-10%
DC bus	310 VDC	Discharged before service
CN2 signal	24 VAC	Call for fan
Ambient	see chart	Record at test



Thermostat Wiring reference

> Use the wiring diagram in Appendix D to confirm connector pinouts before probing.

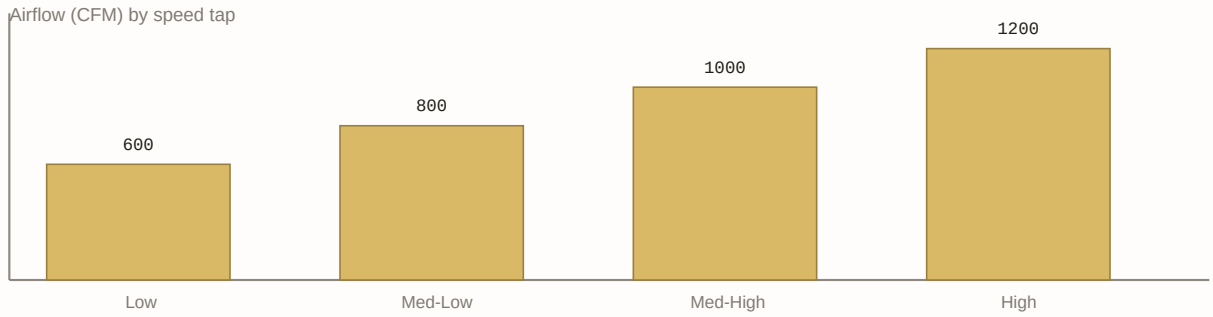
16. Condensate Drainage

Record ambient and supply-air temperatures at the same time so the temperature split can be compared against the charging chart for the current mode.

Procedure notes

- Disconnect and lock out power.
- Confirm 0VAC and 0VDC at the board.
- Inspect connectors for corrosion.
- Record baseline readings before any change.
- Reassemble and re-check operation.

Terminal	Wire	Function
CN2-1	RED	24VAC supply
CN2-2	WHT	Common return
CN2-3	BLU	Tach feedback
CN1-1	BLK	Earth ground
CN1-2	YEL	Thermostat Y



Condensate Drainage reference

> Always confirm 0VAC and 0VDC at the board before making resistance measurements.

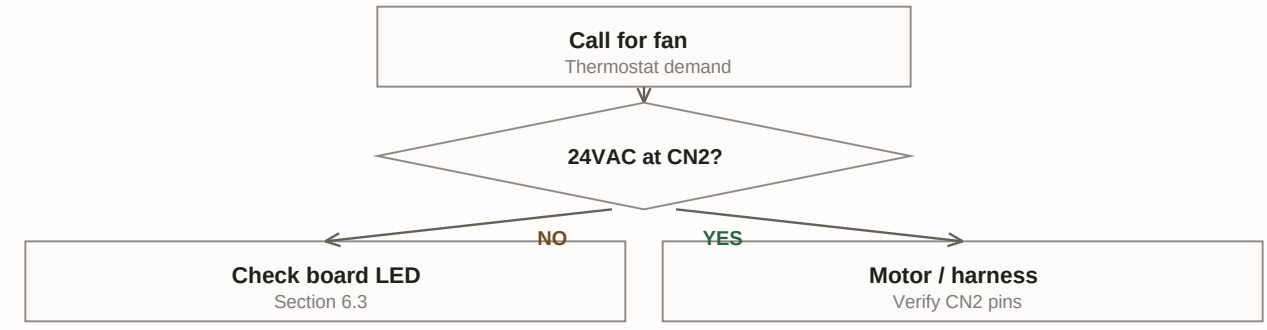
17. Duct Connections

When a reading is out of range, do not replace the component until the associated wiring and connectors have been verified good. Most returned parts test good on the bench.

Procedure notes

- Verify thermostat demand is present.
- Check the low-voltage transformer output.
- Measure the signal at CN2.
- Compare against the table in this section.
- Clear codes only after the repair is verified.

Code	Meaning	Action
E1	High pressure	Check charge and airflow
E2	Low pressure	Check charge and leak
E4	Locked rotor	Check compressor
E5	AC overcurrent	See Section 5.2
E6	Communication	Check harness



Duct Connections reference

> Record the outdoor ambient and the suction line temperature together for an accurate comparison.

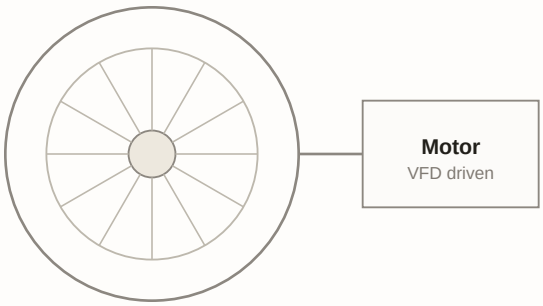
18. Airflow Setup

Torque all electrical terminations to the value shown in the specifications table. Loose terminations overheat and can damage the board terminals over time.

Procedure notes

- Use only factory-approved replacement parts.
- Match the full part number, including revision.
- Transfer configuration jumpers where fitted.
- Confirm harness routing against the reference photos.
- Run a full startup sequence after assembly.

Step	Tool	Spec
Mounting screws	1/4 in driver	2.5 N-m
Terminals	Torque driver	0.8 N-m
Line set	Flare wrench	per chart
Filter	Hand	snug
Panel	1/4 in driver	1.5 N-m



Indoor blower wheel and drive motor
Airflow Setup reference

> If a code returns after clearing, inspect the harness before replacing the board.

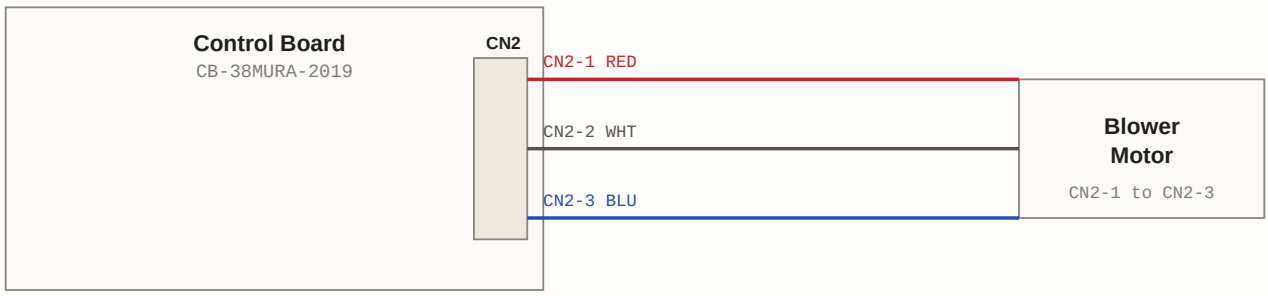
19. Charging the System

Document the final readings and the parts used on the work order. Accurate records shorten the next visit and support the warranty claim.

Procedure notes

- Measure with the unit in the same mode as the fault.
- Allow readings to stabilize for 30 seconds.
- Note the ambient temperature at the time of test.
- Repeat any suspect measurement twice.
- Compare to the published range, not to an assumption.

Speed Tap	CFM	Static
Low	600	0.20 in wc
Med-Low	800	0.30 in wc
Med-High	1000	0.40 in wc
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24VAC control signal from board to blower motor

Charging the System reference

> Use the wiring diagram in Appendix D to confirm connector pinouts before probing.

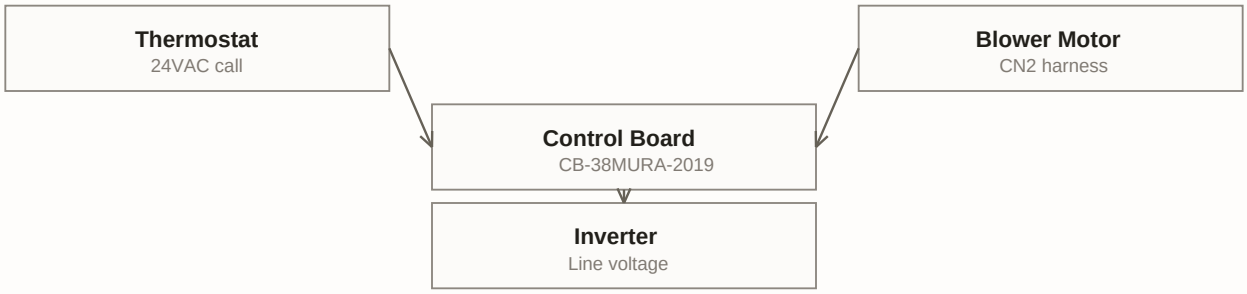
20. System Checkout

This section covers the recommended field procedure for the 2019 Carrier 38MURA. Follow the steps in order and verify each measurement before moving to the next step. Record every reading on the service worksheet.

Procedure notes

- Disconnect and lock out power.
- Confirm 0VAC and 0VDC at the board.
- Inspect connectors for corrosion.
- Record baseline readings before any change.
- Reassemble and re-check operation.

Item	Expected	Notes
Line voltage	208-230 VAC	+/-10%
Low voltage	24 VAC	+/-10%
DC bus	310 VDC	Discharged before service
CN2 signal	24 VAC	Call for fan
Ambient	see chart	Record at test



System Checkout reference

> Always confirm 0VAC and 0VDC at the board before making resistance measurements.

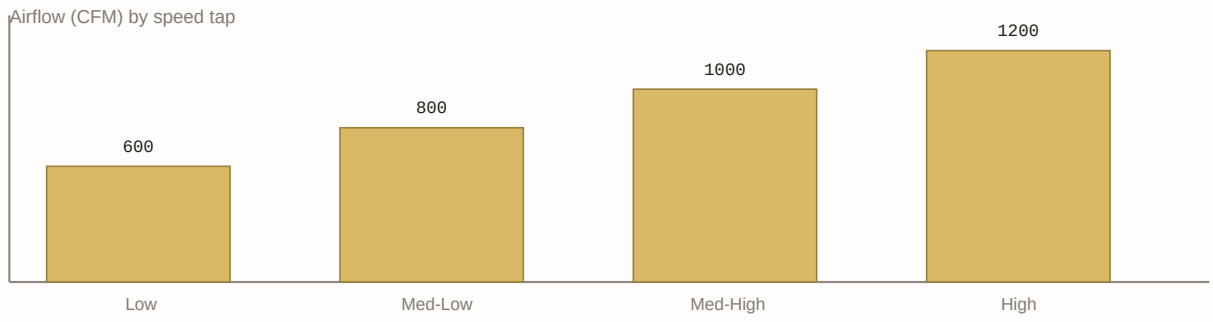
21. Commissioning Checklist

The 38MURA uses a variable-speed inverter to match capacity to load. Because the inverter regulates motor speed, most faults present as a control-signal problem rather than a mechanical failure.

Procedure notes

- Verify thermostat demand is present.
- Check the low-voltage transformer output.
- Measure the signal at CN2.
- Compare against the table in this section.
- Clear codes only after the repair is verified.

Terminal	Wire	Function
CN2-1	RED	24VAC supply
CN2-2	WHT	Common return
CN2-3	BLU	Tach feedback
CN1-1	BLK	Earth ground
CN1-2	YEL	Thermostat Y



Commissioning Checklist reference

> Record the outdoor ambient and the suction line temperature together for an accurate comparison.

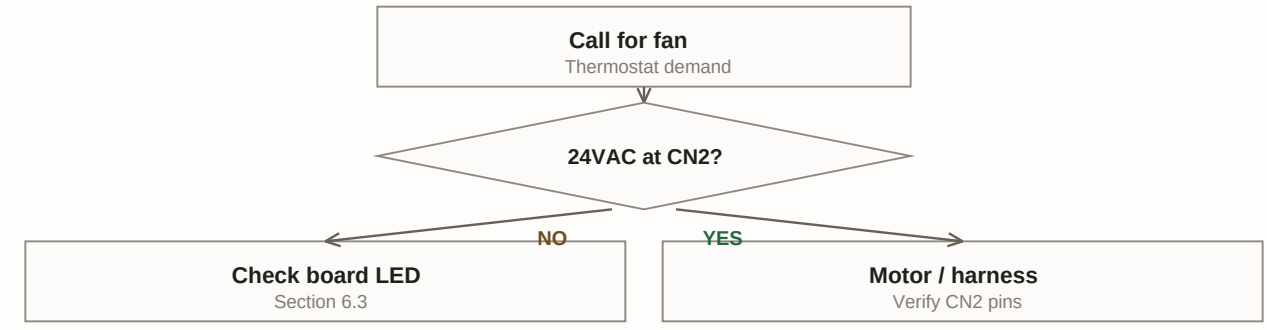
22. Customer Handoff

Before opening any panel, disconnect power at the service switch and wait for the DC bus to discharge. Confirm 0VAC and 0VDC at the board before proceeding.

Procedure notes

- Use only factory-approved replacement parts.
- Match the full part number, including revision.
- Transfer configuration jumpers where fitted.
- Confirm harness routing against the reference photos.
- Run a full startup sequence after assembly.

Code	Meaning	Action
E1	High pressure	Check charge and airflow
E2	Low pressure	Check charge and leak
E4	Locked rotor	Check compressor
E5	AC overcurrent	See Section 5.2
E6	Communication	Check harness



Customer Handoff reference

> If a code returns after clearing, inspect the harness before replacing the board.